PROF. GABRIEL GREENBERG

W6 Modularity

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READINGS Fodor, "Précis of *The Modularity of Mind*" (1985); Fodor, *The Modularity of Mind* (1983), III.5 and IV

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1 Two senses of "smart"

- **Two senses in which a machine could be "smart"/"dumb."**
  - **Smart: computational.** The machine can perform computational inference which is essentially ampliative.
  - **Smart: broadly rational.** The machine is broadly rational and can bring the full set of background knowledge to bear in order to make an optimal adjudication on any question.
  - **Dumb: mere causal mechanism.** The machine is just a causal mechanism that doesn't require representation or inferential computation.
  - **Dumb: isolated.** The machine is cut off from rational considerations and cut off from world knowledge.
- **Reflexes.** Reflexes are both non-computational and cut off from world knowledge.
  - **The blink reflex.** "Suppose that you and I have known each other for many a long year . . . you have come to know perfectly well that under no conceivable circumstances would I stick my finger in your eye. . . . Still, if I jab my finger near enough to your eyes, and fast enough, you'll blink. . . . For this reason the blink

reflex is often produced when sober reflection would show it to be uncalled for; like panther-spotting, it is prepared to trade false positives for speed." (p. 71)

The morality tale

- **Perceptual cognition and rationality.** In contradiction of a reflex view of perception, many mid-century theorists (the "New Look") defaulted to a fully cognitive and rational view of perception.
- **Conflation.** In essence they conflated the two notions of smartness: if something could be optimally inferential, it must also be generally rational.
- **Splitting these apart.** Fodor wants to split these apart.
  - **Input systems.** Perceptual systems (or what he calls input systems) include both perception as we think of it (like visual and auditory perception) and also linguistic parsing. These processes are encapsulated from world knowledge, but perform an abductive inferential computation.
  - "If you now imagine a system that is encapsulated in the way that reflexes are, but also computational in a way that reflexes are not, you will have some idea of what I'm proposing that input systems are like." (p. 72)
  - **Central systems.** Central systems, by contrast, are both open-ended with respect to the kind of knowledge they can work with and perform inferential computations.

2 Informational encapsulation and modules

Official definitions

- **Informational encapsulation.** A system is informationally encapsulated when information available to higher cognition cannot influence the processes in lower cognition.
  - **Fodor's formulation.** "The claim that input systems are informationally encapsulated is equivalent to the claim that the data that can bear on the confirmation of perceptual hypotheses includes, in the general case, considerably less than the organism may know. That is, the confirmation function for input systems does not have access to all of the information that the organism internally represents" (p. 69)
- **Module.** "A module is (inter alia) an informationally encapsulated computational system—an inference-making mechanism whose access to background information

is constrained by general features of cognitive architecture, hence relatively rigidly and relatively permanently constrained. One can conceptualize a module as a special-purpose computer with a proprietary database . . . " (Précis, p. 3)

  - "The informational encapsulation of the input systems is, or so I shall argue, the essence of their modularity." (p. 71)

- **Engineering question:** neural back-projections appear to be necessary for cognitive penetration. Are they sufficient?

Examples that demonstrate modularity

- **Müller-Lyer.** "The apparent difference in length of the Mueller–Lyer figures, for example, doesn't disappear when one learns that the arrows are in fact the same size. It seems to follow that at least *some* perceptual processes are insensitive to at least some of one's beliefs. Very much wanting the Mueller–Lyer illusion to go away doesn't make it disappear either" (Précis, p. 2)



- **Motion perception.** "If you (gently) push your eyeball with your finger (as opposed to moving it in the usual way: by an exercise of the will), you *do* get apparent motion. . . . this explicit information, available to you for (e.g.) report, is *not* available to the analyzer in charge of the perceptual integration of your retinal stimulations." (p. 67)

Breaking down informational encapsulation

- **Three forms of interaction.** There are three ways that a higher-level system's content could interact with a perceptual system's content:

(1) **Input.** It could take control over where the system is getting its input from. This might be control over the actual eyes or attentional system.
   - Pylyshyn argues there's plenty of evidence that our high-level systems can control our attention and the directions of our eyes.

(2) **The internal computational process.** It could have influence on the actual internal computational process of the perceptual system.

(3) **Outputs.** It could interact with the outputs of the perceptual system.
   - Everyone accepts that our knowledge has to interact with the outputs of the perceptual system, since perceptions lead to beliefs and these affect our rational choices.

    - "Input analysis may be informationally encapsulated, but perception surely is not." (p. 73)

- **The real debate.** The real debate is whether the high-level systems might influence the middle stages. (option 2)
  - "The penetrability of a system is, by definition, its susceptibility to top-down effects at stages *prior* to its production of output." (p. 74)

Feedback within a module

- **Backwards influence.** Fodor also points out that there can be feedback influence within a module. Modules don't have to be entirely feedforward internally in order to be modular, so long as they're encapsulated from higher-level knowledge.

- **The phoneme restoration effect.** "You make a tape recording of a word (as it might be, the word 'legislature') and you splice out one of the speech sounds (as it might be, the 's'), which you then replace with a tape recording of a cough." (p. 65)

(1) /legi(cough)lature/ $\Longrightarrow$ /legislature/ with a cough 'in the background' (p. 65)

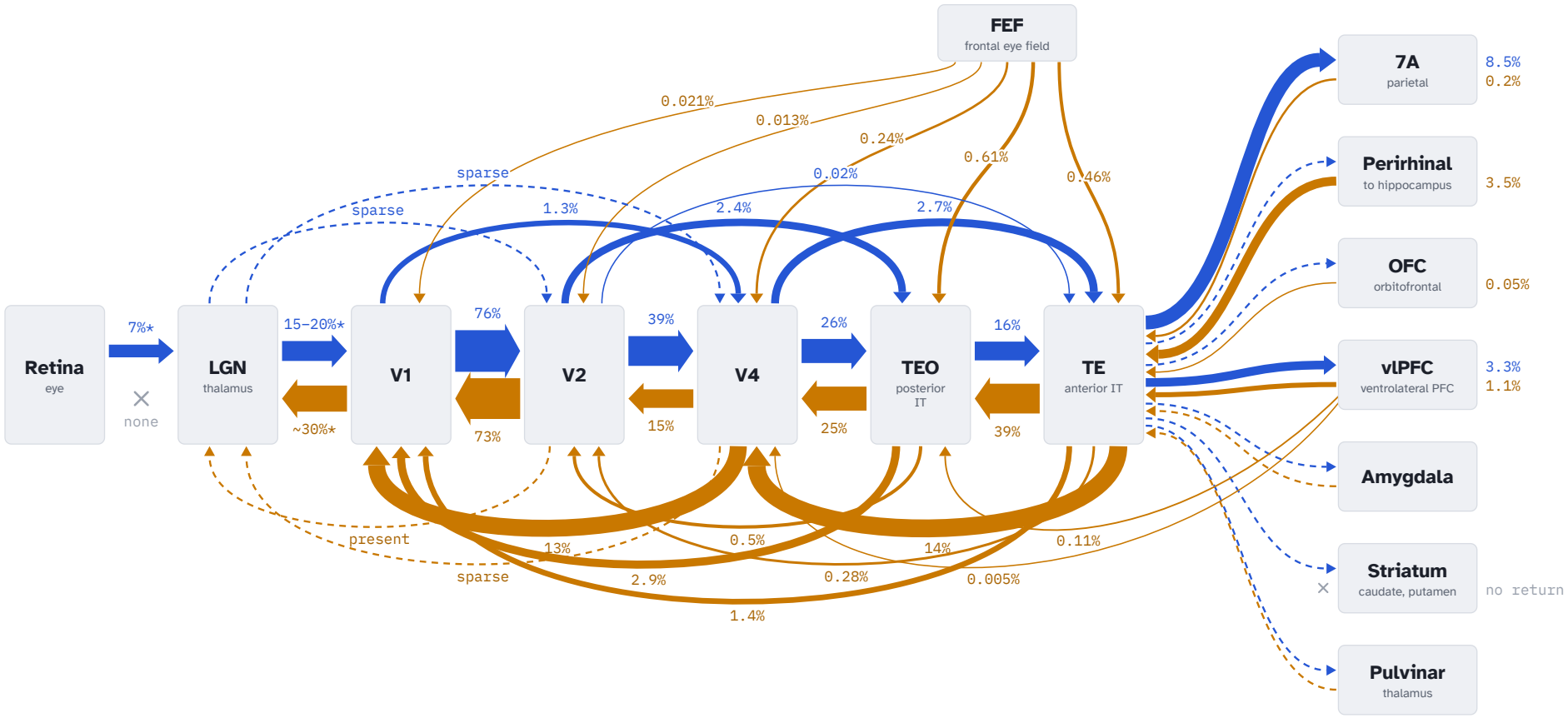
- **Within the language module.** "If, by contrast, the 'background information' deployed in phoneme restoration is simply the hearer's knowledge of the words in his language, then that counts as top-down flow within the language module" (p. 77)

- **Scotomas and the blind spot.** "... visual scotoma (where neurological disorders produce a 'hole' in the subject's visual field). The evidence is that scotoma can mask quite a lot of the visual input without creating a phenomenal blind spot for the subject. What happens is presumably that information about higher-level redundancies is fed back to 'fill in' the missing sensory information. Some such process also presumably accounts for one's inability to 'see' one's retinal blind spot." (p. 65)

Ventral stream: feedforward vs. feedback

Macaque. Arrow width = the sender's share of all the cortical neurons that project to the receiving area. Example: of all neurons elsewhere in cortex that send axons to V1, 73% are in V2.

Feedforward · Feedback · Present, not measured as a share · 0.1% · 1% · 10% · 50%



Footnotes to the ventral stream diagram

- **Fodor's principles.** It vividly demonstrates many of Fodor's principles.
- **Heavy feedback within the system.** You can see there's heavy feedback within the system. In fact feedback projections within the ventral stream outnumber feedforward projections 5:1 or more.
- **From later cognition.** But there are relatively few feedback connections from later cognition to the ventral stream, and virtually none into the middle of it.
- **Suppressed here.** There are other connections that connect to the middle, suppressed here: from the dorsal stream and from areas connected to other modalities.

Visualization generated by Anthropic's Claude AI, with prompting from GG. Data: Markov et al. 2014, *Cerebral Cortex* 24: 17–36 (macaque retrograde tracer counts); LGN values are synapse counts (Van Horn et al. 2000; Garcia-Marin et al. 2019).

3 Side note: mental imagery and encapsulation

- **Mental imagery.** Mental imagery is manifestly influenced by and generated from higher-order cognition. I can give you a verbal instruction to generate a mental image, and you can comply.
- **Visual machinery.** It's also now known that mental images play out over the same neural machinery as perception, albeit to a weaker degree. (This doesn't seem to be a contradiction to the general spirit of informational encapsulation).
- **A module as a process.** If this is not a counterexample to modularity, then a "module" must be understood as a type of process: an informationally encapsulated process, but not really a system in the sense of a subset of a computational architecture. Plainly the computational architecture can be influenced in certain ways by higher cognition.
- **Is "module" the right metaphor for us here?**
  - Do modules map onto computational systems?
  - Do they map onto specific kinds of processes?
  - How do they map onto neural architecture?

4 The rationale for a modular system

- **Bull-headed.** "We now add that input systems are bull-headed and that this, too, makes for speed. . . . speed is purchased for input systems by permitting them to ignore lots of the facts." (p. 70)
- **The epistemic rationale.** We want some systems to encode information in a way that is inflexible to our knowledge or ignorance, and to act more like an automated system.
  - **Don't anther.** "Ogden Nash once offered the following splendidly sane advice: 'If you're called by a panther/don't anther.' Roughly, we want the perceptual identification of panthers to be very fast and to err, if at all, only on the side of false positives." (p. 70)
  - "in the rush and scramble of panther identification, there are many things I know about panthers whose bearing on the likely pantherhood of the present stimulus *I do not wish to have to consider*. As, for example, that my grandmother abhors panthers . . . " (p. 71)
  - **No use to God.** "the *point* of perception is, surely, that it lets us find out how the world is even when the world is some way that we *don't* expect it to be. The teleology of perceptual capacities presupposes a considerably-less-than-omniscient-organism; they'd be no use to God. If you already *know* how things are, why look to *see* how things are?" (pp. 67–68)
- **Fast dumb systems and slow contemplative ones.** "Nature has contrived to have it both ways, to get the best out of fast dumb systems *and* slow contemplative ones, by simply refusing to choose between them." (Précis, p. 4)

5 Modular knowledge and the inverse problem

- **The poverty of the stimulus argument.** What Fodor calls the poverty of the stimulus argument is, for us, the inverse problem.
  - "A Poverty of The Stimulus Argument alleges that there is typically *more information* in a perceptual response than there is in the proximal stimulus that prompts the response; hence perceptual integration must somehow involve the *contribution* of information by the perceiving organism." (Précis, p. 2)
- **Lots of background information.** To solve the inverse problem for vision requires lots of background information.

- **Subdoxastic beliefs?** Fodor calls these subdoxastic beliefs, but I think this is a tough way to put it.
  - "Let us call the epistemic relation that a native speaker has to the grammar of his language *subdoxastic* belief. . . . subdoxastic beliefs are *unconscious*. But, Stich adds, subdoxastic beliefs are also typically 'inferentially unintegrated.' " (p. 84)
- **Better: architecturally encoded information.** This is "information" that is encoded into the neural architecture, which constrains the perceptual system in making certain inferences. This information could be accrued during the lifetime of the organism, but perhaps much more significantly, over the whole evolutionary history of the organism.
- **Information encoded in the system vs. knowledge available to the agent.** Fodor characterizes the information available to the perceptual system (projectable hypotheses) as considerably less than what the organism knows as a whole. I think that's not quite the right way of putting it.
  - "We have repeatedly distinguished between what the input systems compute and what the organism (consciously or subdoxastically) *believes*. Part of the point of this distinction is that input systems, being informationally encapsulated, typically compute representations of the distal layout on the basis of less information about the distal layout than the organism has available." (p. 102)
  - **Difference in kind.** The class of information that's available to support perceptual inference is quite different from what the organism knows, and it might actually extend beyond it quite radically: all this detailed knowledge about the statistical distribution of faces, ambient light, and so on.

6 Central systems and isotropy

The difference between the perceptual systems and the central systems

- **The main quadrant.** There's domain-general and domain-specific, and then encapsulated and non-encapsulated.

	Encapsulated	**Non-encapsulated**
Domain-specific	Perceptual systems	"answers a relatively narrow range of questions but in doing so it uses whatever it knows" (p. 103)
Domain-general	"gives its answers off the top of its head" (p. 104)	Central systems

  - "Roughly, domain specificity has to do with the range of questions for which a device provides answers (the range of inputs for which it computes analyses); whereas encapsulation has to do with the range of information that the device consults in deciding what answers to provide." (p. 103)

- **Perceptual systems.** "these tend to be input driven, very fast, mandatory, superficial, encapsulated from much of the organism's background knowledge, largely organized around bottom-to-top information flow, largely innately specified (hence ontogenetically eccentric), and characteristically associated with specific neuroanatomical mechanisms . . . They tend also to be domain specific" (Précis, p. 4)

- **Central systems.** "these are everything that perception is not: slow, deep, global rather than local, largely under voluntary (or, as one says, 'executive') control, typically associated with diffuse neurological structures, neither bottom-to-top nor top-to-bottom in their modes of processing, but characterized by computations in which information flows every which way. Above all, they are paradigmatically *un*encapsulated" (Précis, p. 4)

	Perceptual systems	**Central systems**
Domain	domain specific	domain neutral
Speed	very fast	slow
Automaticity	mandatory	largely under voluntary ('executive') control
Encapsulation	encapsulated	unencapsulated (isotropic/ Quineian)
Flow of information	bottom-to-top	fully connected
Neural basis	specific neurological mechanisms	diffuse neurological structures
Function	input analysis	fixation of belief

(Précis, p. 4; p. 112)

Motivations for a central system

- **The two features.** Why think there should be a system that has the two features that a central system has?

- **Core features.** It has access to the output from all the input systems; it has access to world knowledge and memory; and it can perform free-ranging inferential computations that relate these sources. Goals would also go along with that.

  - "Central systems look at what the input systems deliver, and they look at what is in memory, and they use this information to constrain the computation of 'best hypotheses' about what the world is like." (p. 104)
  - "However, an interface between perception and utilities must take place *somewhere* if we are to use the information that input systems deliver in order to determine how we ought to act." (pp. 102–103)

- **The isotropy point.** In order to make a truly optimal decision you need to take in all of these factors.

Isotropy

- **Isotropy vs. Quinean.** Confirmation is isotropic if any given hypothesis could be supported (in principle) by any other fact. Confirmation is Quinean if any given hypothesis could be supported by holistic features of the total belief system.
  - **Isotropy.** "By saying that confirmation is isotropic, I mean that the facts relevant to the confirmation of a scientific hypothesis may be drawn from anywhere in the field of previously established empirical (or, of course, demonstrative) truths. Crudely: everything that the scientist knows is, in principle, relevant to determining what else he ought to believe. In principle, our botany constrains our astronomy, if only we could think of ways to make them connect." (p. 105)
  - **Quinean.** "By saying that scientific confirmation is Quineian, I mean that the degree of confirmation assigned to any given hypothesis is sensitive to properties of the entire belief system . . . simplicity, plausibility, and conservatism are properties that theories have in virtue of their relation to the whole structure of scientific beliefs *taken collectively*." (pp. 107–108)

- **Open-ended inference.** Related to Newell's knowledge level: unstructured, infinite.

- **Arbitrarily long-distance connections.** One of the morals of isotropy is that there have to be arbitrarily long-distance connections between different parts of knowledge in order to perform rational inference. Keep that idea in mind as we get into LLMs.
  - "The point about all this is that 'analogical reasoning' would seem to be isotropy in the purest form: a process which depends precisely upon the transfer of information among cognitive domains previously assumed to be mutually irrevelant [sic]. By definition, encapsulated systems do not reason analogically." (p. 107)

– "Quineian/isotropic systems, where more or less any subsystem may want to talk to any other at more or less any time. . . . The point is that in Quineian/isotropic systems, it may be *unstable, instantaneous* connectivity that counts." (p. 118)

- **Causal inference** as an example: why did this happen? Open-ended search for causes.

7 Bayesian learning and the frame problem

Rational decision making

- **Probabilistic reasoning.** If we think about optimal decision making, our model is rational decision making or rational learning in a Bayesian agent.
- **Bayesian update.** Bayesian conditionalization on a novel proposition updates our overall web of probabilities holistically. It's possible because our notion of an optimal rational agent can instantly implement this mathematical function over infinite possibilities.

The frame problem

- **A finite agent with finite time.** The problem is now: how, for a finite agent with finite time, do we do the equivalent of that? This is at least related to what Fodor is calling the frame problem.
  – "the 'frame problem' (i.e., the problem of putting a 'frame' around the set of beliefs that may need to be revised in light of specified newly available information. . . .)" (pp. 112–113)
- **Phoning Mary.** "the robot is presented with the job of phoning Mary and finding out whether she will be late for dinner. . . . Some (but, in principle, not all) such consequences are ones that the robot must be designed to monitor since they are relevant to 'updating' beliefs upon which it may eventually come to act. Well, *which* consequences? . . . Unless the robot can be assured that some of its beliefs are invariant under some of its actions, it will never get to *do* anything." (p. 113)
- **The problem.** The problem that Fodor thinks there's not really a theoretical solution for, or a cognitive scientific solution for, is how to integrate a belief into a belief set. Or, to think of it as computing a function: how to take an existing belief set or cognitive set and a new proposition, and output an appropriate new cognitive set.

$$update(C_{\text{old}}, p) = C_{\text{new}}$$

C_{old} = old cognitive set p = new proposition C_{new} = new cognitive set

  - "Slightly more precisely, the problem is, given an arbitrary belief world W and a new state description 'a is F', what is the appropriate successor belief world W'? . . . it is the general problem of inductive confirmation." (p. 115)

- **The dramaticness of his own conclusion.** Fodor despairs at finding any theoretical solution to this.
  - "the more global (e.g., the more isotropic) a cognitive process is, the less anybody understands it." (p. 107)
  - "I am suggesting that there is a good reason why nothing is known about it—namely, that there is nothing *to* know about it." (p. 119)

- **Contemporary AI.** I'm not sure what Fodor would think of contemporary AI, because here we have a system that solves the frame problem, though not necessarily in the same way we do. But it's not obvious that we have a theory of how these AIs solve the problem.

- **Reflections on the function of thinking.**
  - There's something a little odd about Fodor's despair...
  - Plausibly, the occurrent stream of thought just *is* the process of integrating our beliefs into the web of beliefs.
  - As we think through things, we work through their consequences and further develop the integration of our beliefs into our background commitments.
  - Isotropic integration isn't a process that somehow happens and we have to retroactively figure out how. Arguably, our active offline cognition just is doing exactly that.